

Homework 6

1. Sequential Competitive Equilibrium with taxation

Consider an infinite horizon model populated by mass one of identical consumers, mass one of identical firms, and the government.

In every period, the consumers have one unit of time endowment that they can split between leisure and working for hire. They also own a representative stock market portfolio that pays off dividends (which consumers take as given) in every period. Finally, they own capital stock which they rent to the firm in the beginning of each period. Capital depreciates at rate δ during the production process. The consumers' capital stock holding can be adjusted over time because the consumers decide in every period how much of their total revenue to consume, and how much to save as capital stock for the next period. The consumers' preferences are given by

$$\sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$ and $u(c)$ satisfies all the standard assumptions, including the Inada conditions.

The firms have access to production technology $F(k, n)$ in every period (also satisfying all the standard assumptions, including the Inada conditions), where k and n are capital and labor input, respectively. They can rent capital and hire labor on the competitive markets in order to maximize profits, and distribute all end-of-period profits to their shareholders in the form of dividends.

There are three markets in every period: market for goods (where the prices are normalized to 1), market for labor (where labor services are traded at price w_t), and market for rental capital (where capital is rented at price r_t).

The government needs to finance a stream of exogenous government expenditures $\{G_t\}_{t=0}^{\infty}$ (which it purchases from the goods market) using three types of taxes: lump-sum taxes T_t that apply to all the consumers equally regardless of their actions, proportionate tax s_t on consumption (meaning that if a consumer spends c_t on consumption, $s_t c_t$ is paid to the government as sales tax), and proportionate tax τ_t on capitals gains income (meaning that if a consumer earns $r_t k_t$ in rental income, (s)he needs to pay $\tau_t r_t k_t$ to the government). The government budget constraint is satisfied in every period, i.e.,

$$G_t = T_t + s_t c_t + \tau_t r_t^* k_t^*$$

must hold, where stars denote equilibrium values.

- (a) Define the sequential trade competitive equilibrium in this economy. In the process, define all the necessary variables and decision problems.

- (b) Derive the set of necessary and sufficient conditions that fully characterize the equilibrium.
 - (c) Set up the Social Planner's problem in this economy and derive the set of necessary and sufficient conditions characterizing the efficient allocation.
 - (d) Show that the equilibrium allocation is efficient if and only if $\tau_t = 0$ and $s_t = s_{t+1}$ for all $t \geq 0$.
 - (e) It appears, a constant sales tax rate in this economy is not distortionary. Can you think of a simple (and familiar to you) modification of this model in which any positive consumption tax would be distortionary?
2. Consider a discrete-time, infinite horizon model with two types of agents (mass 1 of each type) having access to production technologies of different goods. Type 1 agent can produce a 'general' good x_t using production technology

$$x_t = A_x k_t^\alpha, \quad \alpha \in (0, 1),$$

where k_t is capital used in production. Type 2 agent can produce a 'specialized' good y_t using technology

$$y_t = A_y k_t^\gamma, \quad \gamma \in (0, 1).$$

In both cases, operational capital fully depreciates in production.

The general good can be either consumed or used as capital in production of either goods, but the specialized goods must be consumed in the period in which it is produced.

In period 0, agents of type 1 and 2 are endowed with capital stocks k_t^1 and k_t^2 , respectively. Agent i 's lifetime utility is given by

$$\sum_{t=0}^{\infty} \beta^t u(x_t^i, y_t^i), \quad \beta \in (0, 1), \quad i = 1, 2$$

where $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ is a strictly increasing, strictly concave, and continuously-differentiable in each argument, such that $\lim_{c \rightarrow 0} u_1(c, \cdot) = \lim_{c \rightarrow 0} u_2(\cdot, c) = +\infty$.

In every period, general and specialized goods are traded on competitive markets; no other markets are open.

- (a) Define the sequential-trade competitive equilibrium in this economy.
- (b) For each type of individual, derive the set of necessary and sufficient conditions that fully characterize their optimal choices.
- (c) Assume that $u(c) = \ln(x) + a \cdot \ln(y)$. Obtain the condition(s) that pin down the equilibrium prices in the steady state equilibrium (containing the price(s) and the model's parameters, no other endogenous variables).

- (d) Your classmate states that it is apparent from the type 2's Euler equation that the steady state amount of capital used in the production of the specialized good increases as A_y increases. Do you agree with your classmate? If not, justify your answer and provide a formal argument explaining how an increase in A_y affects the steady state amount of capital used in the production of the specialized good.
3. Consider a simple modification of the growth model considered in class. Suppose that consumers have no leisure in the utility function, and that the government needs to spend $G > 0$ in every period to keep the economy safe. This government expense is financed through a payroll tax, that is $G = \tau_t \cdot w_t \cdot 1$, where w_t is the equilibrium wage, and τ_t the equilibrium tax rate in period t . Set up the recursive competitive equilibrium in this economy.
4. **Recursive Competitive Equilibrium with heterogeneous firms**
 Consider the following modification of the standard model (without leisure in the utility function) studied in class: Suppose there are two types of firms, equal mass of each type, both produce the same final good but operate different production technologies $F(k)$ and $G(k)$, respectively. There is a continuum of mass $1/2$ of each firms. Assume that the marginal products of capital for each production technology is infinite if capital approaches 0. Set up the recursive competitive equilibrium in this environment.